BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

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Abstract

In same-domain legal consultation archives, dense retrieval often surfaces a relevant client question while failing to return the lawyer's advice turn—an *episode incompleteness* failure that leaves prior recommendations hard to recover under lexical interference from neighbouring matters. BIMS-LEGAL indexes Chinese consultation logs in a dual store: a fast episodic turn index with session identifiers and a slower BIRCH semantic store. Soft O2 binds session siblings by attenuated score inheritance (βs) rather than hard score copy. Exact replay is reported only as a surface-form reference; the main channels are paraphrase, follow-up, and advice-recall. On a shared turn-level index Soft O2 improves over dense FlatIP, with hard hydration and shuffled-*sid* controls separating binding from score copying and membership noise. Gains are largest where a partial episode hit can pull in its answer sibling. Soft O2-C and gated Hybrid extend binding to BIRCH clusters under a Mix protocol with off-session gold; when query and gold already share a session, Soft O2 is preferred. We report Answer Hit, Episode Completeness, graded nDCG with fixed gold-session IDCG, a complete/incomplete/miss failure taxonomy, and paired significance tests.

Keywords: legal information retrieval, consultation memory, soft episode binding, dual-store memory, cluster binding, Chinese legal corpora

1. Introduction

A junior associate may ask a firm assistant: "Last month you advised this client on terminating a fixed-term contract—what remedy did we recommend?" The system returns several near-identical "contract termination"

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questions, ranks the stored user utterance highly, and never surfaces the partner's written advice. The subsequent reply may sound fluent yet be grounded in the wrong episode—or in none. In consultation archives, questions and answers share specialized terminology; neighbouring matters are topically adjacent yet legally distinct; and the professionally useful evidence unit is the full consultation *episode*.

Legal informatics has long studied statutory interpretation, case retrieval, and advisory systems Martinez-Gil (2023); Ashley (2017). Recent legal large language models (LLMs) improve statutory and fact-pattern answering Yue et al. (2023); Huang et al. (2023); Yue et al. (2024). Classical legal information retrieval (IR) mainly targets *external* authority—statutes, cases, or community answers—as documents Askari et al. (2024); Louis et al. (2024); Liu et al. (2022); Zhang et al. (2024). Memory-augmented dialogue systems recover evidence from evolving conversation logs Wu et al. (2024); Maharana et al. (2024); Packer et al. (2023). The gap addressed here is *internal consultation memory*: retrieving prior client-specific advice from a growing same-domain archive under lexical interference.

BIMS-LEGAL (Brain-Inspired Memory System for legal consultation archives) separates a fast episodic pathway—timestamped turn embeddings with speaker role and session identifier *sid*—from a slow semantic pathway based on Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH) with deferred consolidation Zhang et al. (1996). Complementary Learning Systems (CLS) theory McClelland et al. (1995); Kumaran et al. (2016) is used only as an architectural analogy for that dual-pathway split. Under same-domain interference, dense turn retrieval in a vector database often returns a question turn while its answer sibling remains outside top-k—a system-level episodic memory failure we term *episode incompleteness*, distinct from a full session miss and exacerbated by IVFPQ score collapse and context fragmentation at moderate archive sizes.

The *algorithmic contributions* are soft binding operators that complete episodes without hard score copy:

- **Soft O2 (session soft binding).** When a turn is retrieved with score s, siblings sharing *sid* inherit $\max(s_{t'}, \beta \cdot s)$, surfacing answer turns under question-hit / answer-miss while preserving rank competition among candidates.

- **Soft O2-C and gated Hybrid (cluster soft binding).** The same soft-inheritance rule applies within BIRCH clusters when gold evidence is not co-session with the query; Hybrid triggers cluster binding only on direct dense hits to avoid displacing same-session siblings.

Two **implementation operators** support large shared-store experiments but are not standalone algorithmic contributions:

- **O1 (exact FlatIP indexing).** At hundreds-to-thousands of turns, IVFPQ product quantization is under-trained and collapses Answer Hit; FlatIP restores faithful turn scores so Soft O2 inherits uncorrupted sibling signals.
- **O3 (bulk-load consolidation).** Deferred BIRCH and disk writes make archive construction tractable without changing retrieval semantics.

We ask four research questions:

- **RQ1.** Under same-domain interference, how prevalent is episode incompleteness (question hit / answer miss) relative to full session miss?
- **RQ2.** Do Soft O2 and Soft O2-C address episode incompleteness under measured index distortion, and what incremental gains does Soft O2 provide over FlatIP and hard parent hydration?
- **RQ3.** On the *same* LegalEp /LegalMem stores, when is the slow (BIRCH) pathway effective? Does Soft O2-C help under standard same-session gold, and under a fair Mix of same-/cross-session gold?
- **RQ4.** Across CAIL multi-turn and LegalEp paraphrase /advice-recall channels, how large and session-tied are Soft O2 gains, and when do Soft O2-C /Hybrid help beyond Soft O2?

Contributions..

1. A dual-store memory architecture for Chinese legal consultation archives, separating fast episodic turn recovery from slower BIRCH semantic integration under same-domain interference.
2. Soft O2, a turn-level session soft-binding operator that recovers missing answer siblings without hard parent hydration or session-max score copy.

3. A three-candidate score-competition account of Soft O2 using the gap ratio $\rho = s_d/s_h$, with default β=0.98 chosen by validation sweep.
4. Turn-level lexical and fusion controls (BM25-turn, dense+BM25 RRF, cross-encoder) under the same evidence unit as Soft O2, with joint episode indexing treated as a coarser alternative design rather than a turn-binding baseline.
5. Soft O2-C and gated Hybrid on Mix cross-session reconstructions, graded nDCG with fixed gold-session IDCG, a failure taxonomy, and paired significance tests.
6. O1 exact FlatIP and O3 bulk-load consolidation as implementation operators for the shared-store ablations.

Section 2 reviews related work. Section 3 presents BIMS-LEGAL operators and the Soft O2 attenuation analysis. Section 4 details corpora, metrics, and protocols. Section 5 reports results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.1. Legal IR and long-horizon dialogue memory

Legal QA and retrieval systems typically ground generation in statutes, cases, or published advice Louis et al. (2024); Askari et al. (2024); Martinez-Gil (2023); Ashley (2017). Conversational legal case retrieval and event-aware similar-case ranking further show how interaction context and structured legal knowledge shape IR effectiveness Liu et al. (2022); Zhang et al. (2024). Those pipelines optimize access to external authority. Consultation memory retrieval targets a different evidence source: *prior consultation episodes* inside an agent's own archive. Confusing two clients' histories can produce fluent yet professionally inappropriate responses even when statute retrieval is correct. Domain LLMs such as DISC-LawLLM, Lawyer LLaMA, and LawLLM improve statutory and advisory generation Yue et al. (2023); Huang et al. (2023); Yue et al. (2024), but they still need a reliable memory index when prior advice must be recovered from a growing same-domain log.

Legal consultation archives add a further difficulty: high lexical overlap among topically adjacent matters, with the professionally useful evidence unit being the full question–answer episode. This setting motivates *dual-store* memory architectures that keep fast episodic turn retrieval separate from slower semantic clustering, and session-aware soft binding operators that recover answer turns under same-domain interference without collapsing rank competition among unrelated candidates.

2.2. Dual-store memory and episodic retrieval

Dual-store designs distinguish fast episodic encoding from slower semantic integration McClelland et al. (1995); Kumaran et al. (2016); Tulving et al. (1972); Atkinson and Shiffrin (1971). Memory architectures for dialogue agents extend context via hierarchical paging, summary banks, graphs, and neuro-symbolic retrieval Packer et al. (2023); Zhong et al. (2024); Rasmussen et al. (2025); Gutiérrez et al. (2024); Modarressi et al. (2023); Wang et al. (2024); Lewis et al. (2020); Wu et al. (2025), but most treat retrieved units as flat passages rather than *multi-turn consultation episodes* whose answer turn may rank below an unrelated question turn sharing legal vocabulary. LongMemEval and LoCoMo evaluate long-horizon recall and QA over personal conversation logs Wu et al. (2024); Maharana et al. (2024), while broader memory-agent surveys emphasize persistence beyond the context window Wu et al. (2025); Tan et al. (2025). Recent conversational RAG work stresses reuse of historical search evidence across turns Mo et al. (2026). BIMS-LEGAL positions its dual-store contribution at this intersection: episodic turn embeddings with *sid* for same-session recovery (Soft O2), and BIRCH centroids with Soft O2-C for cross-session recovery when query and gold evidence occupy different sessions but share thematic structure Zhang et al. (1996). Under dense same-domain interference, a question turn can occupy top-k while its answer sibling is absent. Parent-document hydration and session-max aggregation recover that sibling by hard score copy; Soft O2 instead injects attenuated inherited scores so episode completion does not erase rank competition among unrelated candidates.

2.3. Session-aware retrieval

Parent-document hydration, session aggregation, and joint session indexing are standard session-aware mechanisms for attaching sibling context to a dense hit. Soft episode binding (Soft O2) inherits $\beta \cdot s$ for siblings and keeps ranking competition among candidates. Parent-document and hierarchical retrieval similarly attach child evidence to a parent unit Callan et al. (1994); Dai et al. (2019); Zaheer et al. (2017); Karpukhin et al. (2020); Soft O2 differs by using attenuated turn-level score propagation rather than hard hydration or atomic session documents. Lexical BM25 Robertson and Zaragoza

(2009), dense passage retrieval Karpukhin et al. (2020), dense–sparse Reciprocal Rank Fusion Cormack et al. (2009), and cross-encoder reranking Nogueira et al. (2019); Chen et al. (2024) provide complementary controls outside session membership. Exact flat indexes and product-quantized ANN backends Johnson et al. (2019) further affect whether sibling binding is applied over faithful turn scores.

Contrast with joint-episode indexing.. An alternative way to keep episode context is to concatenate full question–answer pairs into monolithic episode documents (joint indexing). Joint indexing can prevent episode incompleteness by construction, but it changes the evidence unit from individual turns to merged documents. That coarser unit weakens fine-grained turn matching—for example, isolating a specific advice sentence—and raises index memory and update cost when new turns arrive in an online consultation log. Soft O2 instead keeps turn-level representations and candidate competition, recovering missing siblings on demand via attenuated score inheritance (βs) without replacing the underlying turn index. Table 1 summarizes the session-structure contrasts used in the main grids.

Table 1: Session-aware mechanisms compared in this work.

Mechanism	**Unit**	**Expansion**	**Type**	**Rank**
Parent hydration	Session	Insert siblings	Hard	No
Session-max	Session	Session max	Hard	Same*
Joint episode index	Session doc	Atomic	N/A	N/A
Soft O2	Turn+sid	Soft βs	Soft	Yes
Soft O2-C	Turn+cluster	Soft $\beta_c s$	Soft	Yes

*On the reported corpora, session-max rankings coincide with hard parent hydration under full-sibling expansion; main tables report a single hard-expansion baseline. Soft O2 denotes session soft binding; Soft O2-C denotes cluster soft binding on the BIRCH pathway.

In short, the dual-store split is the system design, and Soft O2 /Soft O2-C are the operators evaluated on Chinese legal consultation archives.

3. Method: BIMS-LEGAL

3.1. Architecture overview

Figure 1 summarizes the pipeline. BIMS-LEGAL maintains complementary episodic and semantic stores over the same consultation archive.

Definition 1 (Episodic and semantic elements)**.** *An* episodic turn *(episodic element) is $e_i = (\mathbf{v}_i, t_i, c_i, \phi_i, sid_i)$, where $\mathbf{v}_i \in \mathbb{R}^d$ is an embedding, t_i a timestamp, c_i conversational context (including speaker role), ϕ_i surface features, and sid_i the consultation session identifier. We reserve* session turns *for the weaker notion of turns that share only sid under a grouping key, without requiring role, timestamp, or other episodic metadata to participate in scoring. Soft O2 binds over session siblings* $\mathrm{Sib}(t) = \{t' : sid_{t'} = sid_t\} \setminus \{t\}$*, but the indexed objects themselves remain episodic turns. A semantic element is $s_j = (\mathbf{c}_j, C_j, \kappa_j, \tau_j)$ with centroid $\mathbf{c}_j$, member set C_j, label κ_j, and formation statistics τ_j. Write* $\mathrm{Clus}(t) = \{t' : cid_{t'} = cid_t\} \setminus \{t\}$ *for BIRCH-cluster siblings.*

Definition 2 (Dual-store retrieval map)**.** *Given query q with embedding $\mathbf{v}_q$, base episodic retrieval returns a scored candidate set $\mathcal{R}_0(q) = \{(t, s_t)\}$ under Eq.* (1)*. Soft O2 produces $\mathcal{R}_{\mathrm{O2}}(q) = \mathsf{SoftBind}(\mathcal{R}_0(q); \mathrm{Sib}, \beta)$ (Section 3.3). Soft O2-C produces $\mathcal{R}_{\mathrm{O2C}}(q) = \mathsf{SoftBind}(\mathcal{R}_0(q); \mathrm{Clus}, \beta_c)$ (Section 3.5). Gated Hybrid composes session binding then cross-session cluster binding (Algorithm 2). The returned list is the top-k of the bound set under the rewritten scores.*

3.2. Base scoring and implementation operators (O1, O3)

For query q with embedding $\mathbf{v}_q$, episodic base retrieval scores a memory turn m by

$$\mathrm{score}(m) = \alpha\, S(\mathbf{v}_q, \mathbf{v}_m) + (1-\alpha)\, R(t_m), \tag{1}$$

where S is cosine similarity on ℓ_2-normalized embeddings and

$$R(t_m) = \max\left(0,\ 1 - \frac{\Delta_t}{30 \text{ days}}\right), \tag{2}$$

with Δ_t the elapsed time between query time and the turn timestamp. Unless stated otherwise, α=0.7. Eq. (1) is the ranking score used throughout the legal grids; Soft O2 /Soft O2-C rewrite sibling scores after this fusion step and do not replace Eq. (1).

O1: Exact FlatIP indexing.. Let $\widehat{S}$ denote the similarity induced by an ANN backend. At hundreds to a few thousand turns, FAISS IVFPQ is often undertrained, so $\|\widehat{S} - S\|$ is large enough to reorder turns and break Soft O2 inheritance. O1 forces $\widehat{S} = S$ via exact FlatIP (inner-product search on

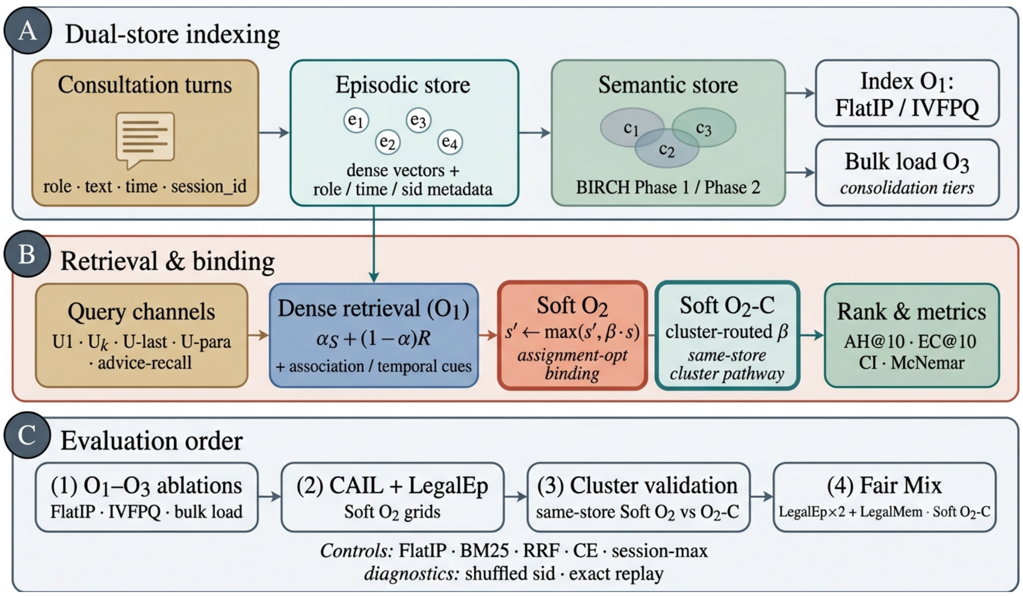

Figure 1: BIMS-LEGAL dual-store pipeline. (A) Episodic turn indexing and BIRCH semantic clustering (O3 bulk-load). (B) Query-time FlatIP retrieval with Soft O2 session binding and Soft O2-C cluster binding; O1 and O3 are implementation operators. (C) Evaluation order: shared-store ablations, Soft O2 grids, and Mix Soft O2-C.

ℓ_2-normalized embeddings). Quantization error depresses Answer Hit (Table A.3: IVFPQ AH 0.330 at M=1600 on LegalEp-DISC exact). O1 is an engineering prerequisite for Soft O2 evaluation, not a standalone retrieval algorithm.

O3: Bulk-load consolidation.. During bulk construction, O3 replaces the per-insert update ($e_i \mapsto \mathsf{BIRCH}(e_i)$; Flush) by a deferred operator $\mathsf{Finalize} = \mathsf{BIRCH}(\{e_i\}_{i=1}^{N}) \circ \mathsf{Flush}$, executed once after all episodic writes. O3 does not change retrieval semantics; it makes $M \geq 400$ shared-store experiments runnable ($\approx$ 11 min wall-clock for M=400; amortized $\approx$ 0.825 s/turn over $\sim$ 800 turns). Quality–latency curves to M=6400 appear in Section 5.6.

3.3. Soft O2: session soft binding

Motivation.. Under same-domain interference, a retrieved question turn can score highly while its answer sibling remains outside top-k—episode incompleteness compounded by IVFPQ score collapse and context fragmentation. Soft O2 completes episodes without hard score copy.

Algorithm 1 Soft O2 session soft binding

Require: Query q, base hits $\mathcal{R}_0$, top-k, attenuation β

```
R ← R_0
for each hit t ∈ R_0 with score s_t do
    for each sibling t′ ∈ Sib(t) absent from R do
        insert (t′, β · s_t) into R
    end for
end for
                                   ▷ Completeness pass after fusion with Eq. (1)
for each session sid represented in R do
    s_max ← max{s_u : u ∈ R, sid_u = sid}
    for each t′ ∈ Sib of sid still absent do
        insert (t′, β · s_max) into R
    end for
end for
return top-k of R by descending score
```

Definition 3 (Soft binding operator)**.** *Given a scored set $\mathcal{R} = \{(t, s_t)\}$, a sibling relation $\mathcal{N}(\cdot)$, and attenuation $\beta \in (0, 1]$, define*

$$\mathsf{SoftBind}(\mathcal{R}; \mathcal{N}, \beta) : \quad s_{t'} \leftarrow \max\big(s_{t'},\ \beta \cdot s_t\big) \quad \textit{for all } t \in \mathcal{R},\ t' \in \mathcal{N}(t). \tag{3}$$

Soft O2 is $\mathsf{SoftBind}(\cdot; \mathrm{Sib}, \beta)$ *with default* β=0.98*. Hard parent hydration is the special case* β=1 *with forced insertion of all siblings; session-max replaces each member score by* $\max_{t \in sid} s_t$*.*

Algorithm 1 states Soft O2 as sibling expansion followed by a completeness pass and top-k truncation, matching the public implementation.

3.4. Soft O2 attenuation and score competition

The attenuation β controls how strongly a retrieved turn promotes its session siblings. Availability of the injected sibling rises with β, while fidelity of the direct hit falls as $\beta \to 1$. Table 2 summarizes the empirical gap-ratio distribution $\rho = s_d/s_h$ on LegalEp stores under an exact-channel query proxy: median ρ lies well below one, and a high-β operating point near 0.98 covers most measured distractors while remaining strictly below hard hydration (β=1). We therefore characterize the availability–ranking trade-off with a three-candidate score-competition model and choose β by validation sweep

Table 2: Gap-ratio statistics on LegalEp M=3000 stores (exact-channel query proxy; strongest measured non-session distractor). Default β is chosen by the AH sweep in Table A.4.

Corpus	ρ_{50}	ρ_{95}	Cover@0.90	Default β
LegalEp-DISC	0.746	0.854	0.981	0.98
LegalEp-Lawyer	0.662	0.815	0.998	0.98

rather than by a closed-form optimum. A zero-margin hinge objective on $\beta \in (0, 1]$ is uninformative for this choice: its rank term vanishes for all $\beta < 1$.

Definition 4 (Three-candidate score-competition model)**.** *For a query whose dense retrieval directly hits a gold turn h with fused score $s_h > 0$, let a be the gold answer sibling and let d be the strongest non-session distractor in a wide candidate pool, with scores s_a and s_d. Soft O2 rewrites*

$$s'_h = s_h, \qquad s'_a = \max(s_a, \beta s_h), \qquad s'_d = s_d, \tag{4}$$

and defines the gap ratio $\rho = s_d/s_h$. Because d is the strongest measured distractor, conditions based on ρ are conservative sufficient conditions for beating that distractor, not exact top-k boundary conditions for every competitor.

The binding analysis is most relevant in the soft-injection regime $s_a \le \beta s_h$, where $s'_a = \beta s_h$.

Proposition 1 (Feasible attenuation band)**.** *Under Definition 4, assume the soft-injection case $s_a \le \beta s_h$. Then (i) availability against the measured distractor requires $\beta > \rho$; (ii) rank preservation of the direct hit over the injected sibling requires $\beta < 1$, and more generally $s'_a < s_h$; (iii) when $\rho < 1$, any $\beta \in (\rho, 1)$ satisfies both conditions in the active injection case. If $s_a > s_h$ already, rank preservation depends on the observed sibling score rather than on β alone.*

Proof. If $s_a \le \beta s_h$, then $s'_a = \beta s_h$. Availability follows from $\beta s_h > s_d$, i.e. $\beta > \rho$. Rank preservation requires $s'_a < s_h$, which reduces to $\beta < 1$ when $s_h > 0$. □

Soft-rank visualization and sweep.. On a validation grid we also plot the soft pairwise surrogate

$$\mathcal{L}_{\mathrm{sr}}(\beta;\rho) = -\log\sigma\big(\tau(\beta-\rho)\big) - \gamma\log\sigma\big(\tau(1-\beta)\big), \tag{5}$$

with logistic σ, temperature τ, and rank weight γ. The curve is a visualization of the availability–ranking trade-off under the measured ρ pool; it is not used as a fitted optimum. Default β=0.98 is taken from the AH sweep in Table A.4, where $\{0.90, 0.95, 0.98\}$ form a stable high-AH band on CAIL follow-up and LegalEp paraphrase as well as exact grids. Advice-recall uses the same default without a separate gap-ratio fit.

Fair controls include hard parent hydration, session-max aggregation, and shuffled *sid*. On two-turn LegalEp episodes, hard hydration and session-max coincide under full-sibling expansion, so tables report a single hard-expansion baseline. Concatenating a full question–answer episode into one retrieval unit (BM25-joint or dense joint-episode documents) changes the evidence unit from turns to documents; we treat that design as an alternative indexing choice rather than a Soft O2 turn-binding baseline (Section 4).

3.5. Semantic consolidation and Soft O2-C

Semantic consolidation follows a BIRCH-style dual-phase procedure Zhang et al. (1996).

Definition 5 (BIRCH dual-phase consolidation)**.** *Let* $\{\mathbf{c}_j\}$ *be current centroids and* θ_H, θ_M *the assign /merge thresholds.* ***Phase 1 (assign).*** *For a new embedding* $\mathbf{v}$*, set* $j^\star = \arg\max_j S(\mathbf{v}, \mathbf{c}_j)$*; if* $S(\mathbf{v}, \mathbf{c}_{j^\star}) \geq \theta_H$ *then* $C_{j^\star} \leftarrow C_{j^\star} \cup \{\mathbf{v}\}$ *and refresh* $\mathbf{c}_{j^\star}$*, else spawn a new cluster.* ***Phase 2 (merge).*** *Periodically, merge pairs with* $S(\mathbf{c}_i, \mathbf{c}_j) \geq \theta_M$ *and purge near-duplicates above a redundancy threshold. Under O3, both phases run inside* Finalize*. Each turn* t *receives a cluster id* cid_t*, inducing* $\mathrm{Clus}(t)$ *in Definition 1.*

Definition 6 (Soft O2-C)**.** *Soft O2-C is* $\mathsf{SoftBind}(\cdot; \mathrm{Clus}, \beta_c)$ *with default* β_c=0.95 $< \beta$*, reflecting noisier thematic neighbourhoods than session membership. A per-cluster sibling budget* B_c *caps* $|\mathrm{Clus}(t) \cap \mathrm{Injected}|$ *so large thematic clusters cannot flood top-*k*.*

Gated Hybrid (Algorithm 2) injects only *cross-session* cluster siblings relative to sessions already covered by dense /Soft O2 hits, and triggers

Algorithm 2 Gated Hybrid Soft O2 + Soft O2-C

Require: Base hits $\mathcal{R}_0$, budgets (k, B_c), attenuations (β, β_c)

$\mathcal{R} \leftarrow \mathsf{SoftBind}(\mathcal{R}_0; \mathrm{Sib}, \beta)$ ▷ session pathway first
$\mathcal{S}_{\mathrm{cov}} \leftarrow \{sid_t : t \in \text{top-}k(\mathcal{R})\}$
$\mathcal{T}_{\mathrm{trig}} \leftarrow \{t \in \mathcal{R}_0\}$ ▷ direct dense hits only
for each trigger $t \in \mathcal{T}_{\mathrm{trig}}$ **do**
  $s_{\max} \leftarrow \max\{s_u : u \in \mathcal{R},\ cid_u = cid_t\}$
  for each $t' \in \mathrm{Clus}(t)$ with $sid_{t'} \notin \mathcal{S}_{\mathrm{cov}}$, up to budget B_c **do**
    $s_{t'} \leftarrow \max(s_{t'}, \beta_c \cdot s_{\max})$
  end for
end for
return top-k of $\mathcal{R}$

cluster binding only on direct dense hits. This prevents thematic confusers from displacing same-session siblings already recovered by Soft O2. Soft O2 applies when query and gold share *sid*. Soft O2-C is the cluster-pathway operator when gold evidence lies in another session of the same thematic cluster.

4. Corpora and Evaluation Protocol

Figure 2 sketches the shared-corpus design used throughout the legal evaluation. Gold needles and same-domain distractors occupy one store. Each query probes that shared archive under fixed interference conditions. Privately rebuilt haystacks are avoided so that system comparisons isolate the retrieval operator.

4.1. Datasets and query channels

Multi-turn authenticity is evaluated on CAIL2024 consultation dialogues. Gold needles are authentic preliminary and final dialogues, with conversation turns completed by label turns where applicable. Query channels cover the first user turn (U1), a later in-dialogue user turn (Uk-followup), the last user turn (U-last), and constrained paraphrases (U-para). Evidence defaults to assistant turns in the target session.

LegalEp-DISC and LegalEp-Lawyer rebuild public DISC-Law-SFT and Lawyer-LLaMA subsets into consultation-episode corpora. Each episode is a natural (question, answer) pair with session identifier *sid*. Reconstruction

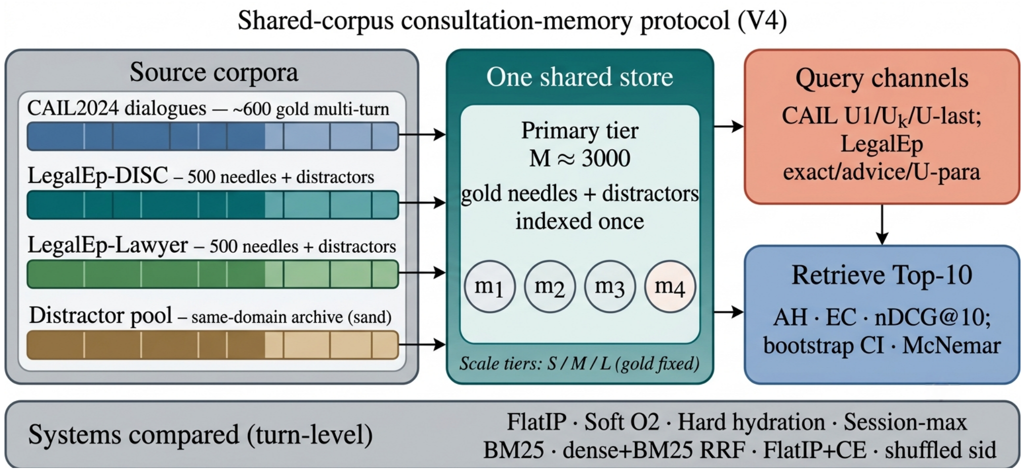

Figure 2: Shared-corpus protocol. Gold needles and same-domain distractors share one store; queries probe multiple channels with turn-level systems and significance tests. Exact replay is diagnostic; paraphrase, follow-up, and advice-recall are primary channels.

applies length bounds and legal-content filters, excludes exam prompts, removes near-duplicates, separates needles from distractors under a fixed seed, and retains an audit subsample. After filtering, each LegalEp corpus retains 3500 passed sessions (500 needles, 2500 distractors at the M=3000 primary tier).

Query-channel policy.. Exact replay uses the stored user question and is reported only as a *diagnostic* upper bound on surface-form recovery. Primary claims use paraphrase, Uk-followup /U-last, and advice-recall queries whose surface form is not identical to the stored question text; paraphrase and advice-recall templates are generated under a fixed seed and audited on a subsample. LegalMem-MT reuses CAIL multi-turn gold under the same store for same-session Soft O2 vs Soft O2-C contrasts and for Mix reconstructions.

Archive size is varied with gold needles held fixed. Unless stated otherwise, Soft O2 transfer tables use the $M \approx 3000$ tier; O1–O2 ablations use M=400, S=300.

4.2. Metrics and baselines

Primary metrics are Answer Hit@k (AH), Episode Completeness@k (EC), and normalized Discounted Cumulative Gain@k (nDCG) at k=10. AH is binary: whether a gold answer turn appears in the top-k. EC is the mean coverage of gold turns within the target session. Session Hit@k is reported only for diagnosis. For nDCG we use fixed graded relevance over the full gold session: answer turns rel=1.0, other same-session turns rel=0.5, all other turns rel=0.0. The ideal denominator (IDCG) is computed from all gold-relevant turns in the target session and truncated at k, so every system on a query shares the same IDCG. Answer Hit and Episode Completeness are the main Soft O2 outcomes; nDCG is reported with the fixed gold-session IDCG so that systems remain comparable on each query. Hard expansion can raise nDCG when copied siblings occupy early ranks, so AH and EC are read together with nDCG.

The primary hypothesis family is Soft O2 vs FlatIP on the nine CAIL and LegalEp transfer channels; for that family we report bootstrap 95% CIs, McNemar mid-p, and Holm-adjusted p-values (Table A.2). Hard-hydration, CE, and Soft O2-C /Hybrid Mix contrasts are secondary.

Turn-level dense FlatIP (O1) is the base dense retriever. Soft O2 applies soft sibling score inheritance on the episodic pathway; Soft O2-C applies the same rule within BIRCH clusters. Hard parent hydration expands a hit to all siblings by copying the trigger score. Lexical controls include BM25 over turns and dense+BM25 Reciprocal Rank Fusion (RRF) under the same turn-level evidence unit. BM25-joint and dense joint-episode indexes retrieve pre-concatenated episode documents and therefore answer a coarser retrieval problem; they are discussed as alternative designs in Section 6.3 rather than Soft O2 turn-binding baselines. A cross-encoder (CE) control reranks the top-30 FlatIP candidates with `BAAI/bge-reranker-v2-m3`. Shuffled *sid*, a β sweep, and IVFPQ provide membership, attenuation, and quantization controls.

4.3. Fair Mix protocol for Soft O2-C

On fully same-session LegalMem grids Soft O2 already recovers gold answers that share *sid*, leaving Soft O2-C little exclusive headroom and exposing it to thematic confusers. To study off-session recovery we rebuild the *same* LegalEp and LegalMem archives under a Mix protocol: 70% of gold episodes are split into query-side S_a and evidence-side S_b with distinct session identifiers, while 30% remain intact. Distractors are unchanged. FlatIP, Soft O2, Soft O2-C, and gated Hybrid share unrestricted dense retrieval on this store; we report overall AH and stratum AH on `cross_session` vs `same_session` queries.

Cluster co-membership and purity under Mix.. After BIRCH consolidation, each cross-session Sa/Sb pair is assigned a shared cluster identifier. This construction uses split metadata available only when building the evaluation store; it does not inject labels at query time. The intent is to test Soft O2-C when thematic co-membership is present. Cluster purity still matters: a shared *cid* that also contains many thematically adjacent distractors narrows Soft O2-C's effective margin, which is why Hybrid gates cluster injection to direct dense hits and caps the per-cluster budget B_c. If recovery still fails under guaranteed Sa/Sb co-membership, the failure is attributable to binding rather than accidental cluster separation. Natural same-cluster rates without this construction are lower (Section 6.3). Soft O2's session membership rule is unchanged. LegalEp Mix uses advice-recall queries; LegalMem Mix uses mid-split Uk-followup queries. LongMemEval /LoCoMo numbers from prior runs on this codebase (QA correctness 70.7% / 68.2%) supply long-horizon memory context only Wu et al. (2024); Maharana et al. (2024).

5. Experiments

Unless noted, all dense retrieval grids use the multilingual encoder `nomic-embed-text-v2-moe` Nussbaum et al. (2025) (768-d) as a single shared embedding backend, with k=10 and seed 42 on the main $M{\approx}3000$ stores. We keep the encoder fixed so Soft O2 vs FlatIP contrasts isolate binding rather than representation changes; sensitivity to alternative Chinese or multilingual encoders (e.g., `bge-large-zh-v1.5`) is discussed in Section 6.3. A five-seed check on a smaller shared store ($M{=}400$) is reported later. Primary contrasts appear in Figures 3–5; full numeric grids are in the appendix. Each main table maps to a released JSON artifact in the accompanying repository.

5.1. Implementation operators and Soft O2 ablation

Table 3 reports the shared-store ablation on DISC-Law and Lawyer-LLaMA ($M{=}400$, $S{=}300$). IVFPQ baseline Answer Hit is 0.540 / 0.397. O1 alone yields a small gain on DISC (0.567) and a near-tie on Lawyer (0.397), confirming that exact indexing is a necessary implementation prerequisite but insufficient alone. O1+Soft O2 raises Answer Hit to 0.780 / 0.680 (+0.240 / +0.283 over baseline) and Episode Completeness to 0.888 / 0.840. O3 is enabled for all configurations in this ladder as a construction prerequisite; it does not alter retrieval scores.

Table 3: O1+Soft O2 ablation on the shared store (M=400, S=300, k=10). EC denotes Episode Completeness (gold-turn coverage); AH denotes Answer Hit. O3 bulk-load is on for all rows. Boldface: best AH and EC within each corpus. Graded nDCG for the $M\approx3000$ grids appears in Table 6.

Corpus	**System**	**EC@10**	**AH@10**
DISC-Law	IVFPQ baseline	0.770	0.540
	+ O1 FlatIP	0.782	0.567
	+ O1+O2 Soft O2	**0.888**	**0.780**
Lawyer-LLaMA	IVFPQ baseline	0.698	0.397
	+ O1 FlatIP	0.698	0.397
	+ O1+O2 Soft O2	**0.840**	**0.680**

5.2. RQ1 failure taxonomy

We classify each query into *complete* (answer evidence in top-k), *incomplete* (target session partially retrieved but answer missing), or *session miss* (no target-session turn in top-k). Table 4 contrasts FlatIP, Soft O2, hard hydration, and shuffled *sid* on primary non-exact channels. Soft O2 converts incompleteness into complete retrieval while session-miss stays nearly fixed; shuffled *sid* restores high incompleteness, so the gain is session-tied rather than generic score inflation. Hard hydration can also drive incompleteness near zero by copying trigger scores, but it is not Soft O2's binding rule and can over-promote siblings (Section 5.4).

5.3. Turn-level Soft O2 controls

Primary Soft O2 claims use a shared turn-level index. Table 5 reports Answer Hit for FlatIP, Soft O2, hard hydration, and shuffled *sid* on the primary Soft O2 transfer channels. Soft O2 beats FlatIP on every non-exact channel; shuffled *sid* collapses toward or below FlatIP; hard hydration is competitive on CAIL AH but Soft O2 remains stronger on LegalEp paraphrase /advice and retains higher EC in the full grids (Tables A.5–A.7). BM25-turn is a same-granularity lexical control (Table A.8): Soft O2 exceeds it on CAIL Uk /U-last (0.698/0.762 vs 0.482/0.413). Joint episode / document-level indexes change evidence granularity and are not Soft O2 protocol competitors.

5.4. Soft O2 on session (CAIL and LegalEp)

Figure 3 and Table 5 summarize AH@10 on authentic CAIL multi-turn channels at $M \approx 3000$ under O1 FlatIP. Soft O2 lifts Answer Hit over FlatIP

Table 4: RQ1 failure taxonomy (k=10). Boldface: best complete / lowest incomplete within each corpus–channel block among FlatIP, Soft O2, and shuffled *sid* (hard hydration shown as a score-copy envelope). Soft O2 cuts incompleteness without raising session miss; shuffled *sid* undoes the gain.

Corpus / channel	System	Complete	Incomplete	Session miss
DISC / u-para	FlatIP	70.8%	26.8%	1.6%
	Soft O2	**89.1%**	**8.7%**	1.8%
	Hard hydr.	97.2%	0.0%	2.8%
	Shuffled O2	66.4%	31.4%	2.2%
DISC / advice	FlatIP	48.0%	17.0%	35.0%
	Soft O2	**60.8%**	**3.8%**	35.4%
	Hard hydr.	65.0%	0.0%	35.0%
	Shuffled O2	45.6%	19.2%	35.2%
Lawyer / u-para	FlatIP	71.5%	25.3%	0.8%
	Soft O2	**94.0%**	**3.8%**	1.6%
	Hard hydr.	98.0%	0.0%	2.0%
	Shuffled O2	72.9%	25.5%	1.6%
CAIL / Uk	FlatIP	45.8%	54.0%	0.2%
	Soft O2	**92.8%**	**7.0%**	0.2%
	Hard hydr.	98.3%	1.5%	0.2%
	Shuffled O2	34.7%	65.2%	0.2%

on every channel (U1 0.788 vs 0.570**; Uk 0.698 vs 0.270**; U-last 0.762 vs 0.245**). Shuffled *sid* removes the gain (U1 0.452; Uk 0.230; U-last 0.200), tying the improvement to correct episode membership (RQ1–RQ2). Hard expansion can raise AH slightly above Soft O2 (e.g., U1 0.802 vs 0.788) while lowering EC (U1 0.578 vs Soft O2 0.664). Soft O2 therefore behaves as a rank-preserving binder: siblings enter under attenuated scores, and episode coverage remains higher than under hard score copy. Table 6 shows the Soft O2>FlatIP ordering under graded nDCG with gold-session IDCG, while incompleteness falls sharply (Table 4). Full numeric grids appear in Tables A.5–A.7.

Figure 4 reports LegalEp transfer on the same AH grids. On LegalEp-DISC exact replay Soft O2 and FlatIP stay near-tied (0.638 vs 0.640), where both turns are already recoverable for many queries—consistent with treating exact as diagnostic. On Lawyer exact Soft O2 improves AH to 0.752 from FlatIP 0.692 (McNemar mid-p=0.024*). Paraphrase yields the clearest

Table 5: Primary Soft O2 controls (AH@10, $M{\approx}3000$). Boldface: best AH within each row among FlatIP / Soft O2 / Hard / Shuffled. Soft O2 is the session-binding claim; shuffled *sid* tests membership dependence; hard hydration is score-copy expansion.

Corpus / channel	FlatIP	Soft O2	Hard	Shuffled
CAIL / U1	0.570	0.788	**0.802**	0.452
CAIL / Uk	0.270	0.698	**0.720**	0.230
CAIL / U-last	0.245	0.762	**0.790**	0.200
LegalEp-DISC / u-para	0.547	**0.648**	0.592	0.469
LegalEp-DISC / advice	0.342	**0.428**	0.402	0.310
LegalEp-Lawyer / u-para	0.568	**0.759**	0.596	0.592
LegalEp-Lawyer / advice	0.332	**0.472**	0.394	0.348

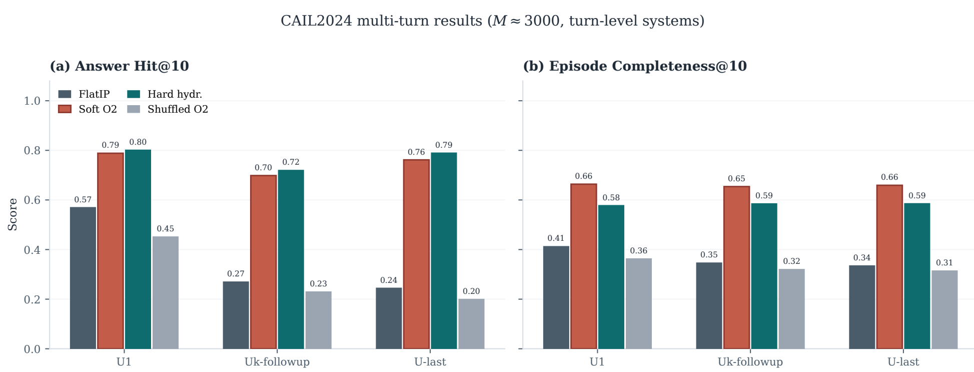

Figure 3: CAIL2024 multi-turn results at tier $M \approx 3000$. Soft O2 improves AH and EC over FlatIP; shuffled *sid* removes the gain; hard expansion can maximize AH at a modest EC cost.

LegalEp gains: Soft O2 reaches 0.648 vs FlatIP 0.547 on DISC and 0.759 vs 0.568 on Lawyer, exceeding hard expansion on both corpora (mid-p<0.001** on both corpora). Advice-recall queries omit the stored question surface form: Soft O2 raises AH to 0.428 from FlatIP 0.342 on DISC and to 0.472 from 0.332 on Lawyer, again above hard expansion (mid-p<0.001**), while shuffled *sid* falls toward or below FlatIP. Graded nDCG mirrors the Soft O2>FlatIP ordering on paraphrase and advice-recall (Table 6). Detailed grids are in Tables A.6–A.7.

Cross-encoder reranking is a strong neural control at the same turn-level evidence unit: FlatIP+CE reranks the top-30 FlatIP candidates with `bge-reranker-v2-m3`. Table 7 shows that FlatIP+CE is competitive—and on LegalEp-DISC exact, strongest—when surface-form overlap already places

Table 6: Graded nDCG@10 with gold-session IDCG (answer *rel*=1.0, other same-session turns *rel*=0.5) and incompleteness rates on primary non-exact channels. Boldface: Soft O2 better than FlatIP within each channel on nDCG. Soft O2 also exceeds shuffled *sid*.

Corpus	Channel	System	nDCG@10	Incomplete
CAIL	Uk	FlatIP	0.445	54.0%
CAIL	Uk	Soft O2	**0.793**	7.0%
CAIL	Uk	Shuffled O2	0.369	65.2%
CAIL	U-last	FlatIP	0.420	57.3%
CAIL	U-last	Soft O2	**0.791**	5.3%
CAIL	U-last	Shuffled O2	0.348	71.2%
LegalEp-DISC	u-para	FlatIP	0.681	26.8%
LegalEp-DISC	u-para	Soft O2	**0.759**	8.7%
LegalEp-DISC	u-para	Shuffled O2	0.627	31.4%
LegalEp-DISC	advice	FlatIP	0.466	17.0%
LegalEp-DISC	advice	Soft O2	**0.524**	3.8%
LegalEp-DISC	advice	Shuffled O2	0.436	19.2%
LegalEp-Lawyer	u-para	FlatIP	0.744	25.3%
LegalEp-Lawyer	u-para	Soft O2	**0.844**	3.8%
LegalEp-Lawyer	u-para	Shuffled O2	0.705	25.5%
LegalEp-Lawyer	advice	FlatIP	0.483	23.4%
LegalEp-Lawyer	advice	Soft O2	**0.582**	2.2%
LegalEp-Lawyer	advice	Shuffled O2	0.454	25.0%

both question and answer near the top of the dense pool (DISC exact AH 0.722 for FlatIP+CE vs 0.638 for Soft O2). On channels with semantic shift or multi-turn drift, Soft O2 pulls ahead: CAIL Uk 0.698 vs FlatIP+CE 0.375, CAIL U-last 0.762 vs 0.342, and LegalEp-Lawyer exact 0.752 vs 0.708. A structural limit of the reranker explains the split: CE can refine ranking among candidates already retrieved by FlatIP, but it cannot promote an answer sibling that never entered the top-30 pool. Soft O2 recovers such siblings by propagating attenuated scores through session links, which helps precisely when the stored advice turn has weak direct lexical overlap with the query (paraphrase and advice-recall; cf. Tables 5 and 6).

5.5. Soft O2-C and gated Hybrid under Mix

When gold answers already share *sid* with the query, Soft O2 is sufficient and cluster binding is unnecessary. Table 8 asks the off-session question under the Mix protocol of Section 4.3: 70% cross-session Sa/Sb gold and 30% intact same-session gold, with FlatIP, Soft O2, Soft O2-C, and gated

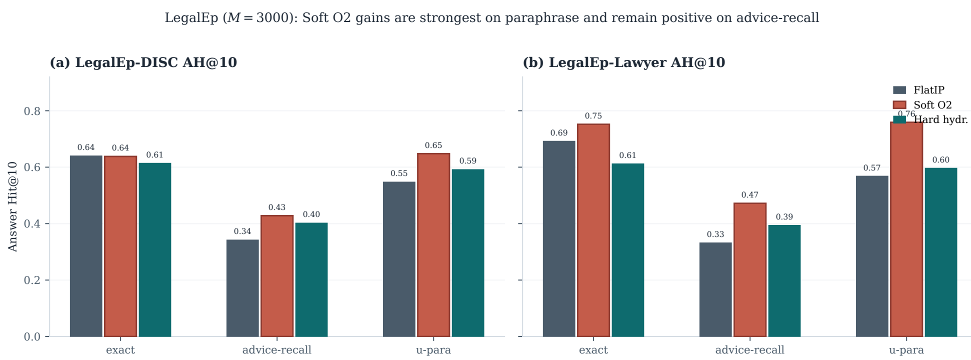

Figure 4: LegalEp AH@10 (M=3000, 500 needles). Soft O2 gains are strongest on paraphrase and remain positive on advice-recall; exact transfer is corpus-dependent and treated as diagnostic.

Table 7: Cross-encoder control (AH@10). FlatIP+CE reranks the top-30 FlatIP candidates with `bge-reranker-v2-m3`. Boldface: best AH within each row.

Corpus / channel	FlatIP	FlatIP+CE	Soft O2
CAIL / U1	0.585	0.688	**0.788**
CAIL / Uk	0.265	0.375	**0.698**
CAIL / U-last	0.228	0.342	**0.762**
LegalEp-DISC / exact	0.572	**0.722**	0.638
LegalEp-Lawyer / exact	0.636	0.708	**0.752**

Hybrid sharing unrestricted dense retrieval.

On the cross-session stratum Soft O2-C improves over Soft O2 on LegalEp-Lawyer (0.491 vs 0.446) and LegalMem-MT (0.537 vs 0.503): cluster binding can surface split-episode gold that session membership alone misses. Gated Hybrid converts that mechanism into an overall gain on LegalEp-Lawyer (0.534 vs Soft O2 0.496; mid-p=0.010*) and a directional gain on LegalMem-MT (0.646 vs 0.618), while preserving most same-session hits (0.560 / 0.867). On LegalEp-DISC the Hybrid edge is negligible (0.486 vs 0.482). Ungated Soft O2-C can raise cross-session hits yet displace intact same-session evidence, which is why Hybrid triggers cluster binding only on direct dense hits. Same-session Soft O2 vs Soft O2-C contrasts remain Soft O2-led (Appendix Appendix A, Table A.1).

McNemar mid-p on paired Answer Hit; * $p<0.05$, ** $p<0.01$. Cluster co-membership of Sa/Sb pairs follows the Mix protocol in Section 4.3.

Table 8: Mix evaluation of Soft O2-C and gated Hybrid (same store, unrestricted dense; AH@10). Gold is 70% cross-session Sa/Sb + 30% intact same-session. Boldface: best overall AH and best cross-session AH within each corpus. Significance vs Soft O2: * $p<0.05$, ** $p<0.01$.

Corpus	**System**	**AH**	**AH$_{\text{cross}}$**	**AH$_{\text{same}}$**	**mid-p vs Soft O2**
LegalEp-DISC Mix	FlatIP	0.478	0.477	0.480	—
	Soft O2	0.482	0.443	**0.573**	—
	Soft O2-C	0.410	0.443	0.333	0.0004**
	Hybrid	**0.486**	**0.469**	0.527	0.8099
LegalEp-Lawyer Mix	FlatIP	0.462	0.454	0.480	—
	Soft O2	0.496	0.446	**0.613**	—
	Soft O2-C	0.464	0.491	0.400	0.0857
	Hybrid	**0.534***	**0.523**	0.560	0.0105*
LegalMem-MT Mix	FlatIP	0.534	0.526	0.553	—
	Soft O2	0.618	0.503	**0.887**	—
	Soft O2-C	0.500	0.537	0.413	0.0000**
	Hybrid	**0.646**	**0.551**	0.867	0.1837

5.6. Scale behaviour

Figure 5 and Table A.3 report quality and warm-query latency on LegalEp-DISC exact (Q=100, three repeats; $M \in \{100, 400, 1600, 6400\}$). Soft O2 retains an AH advantage at each tier; IVFPQ collapses at M=1600 (AH 0.330), motivating O1; absolute latency grows with M. We treat O1 FlatIP as an engineering prerequisite for Soft O2 at these scales and avoid claiming deployment readiness beyond the measured curve.

End-to-end QA audit (summary).. A dual-protocol QA audit (N=270 per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`) reports judged answerability under Soft O2 retrieval alone; it is not a paired FlatIP vs Soft O2 generation experiment. Full protocol and stratified results appear in Section 6.2. On a smaller shared store (M=400, five seeds), Soft O2 AH on LegalEp-DISC is 0.851 ± 0.008 versus FlatIP 0.698 ± 0.040.

6. Discussion

6.1. Findings and scope

RQ1–RQ2 show that episode incompleteness is a substantial FlatIP failure mode under same-domain interference, and that Soft O2 converts many

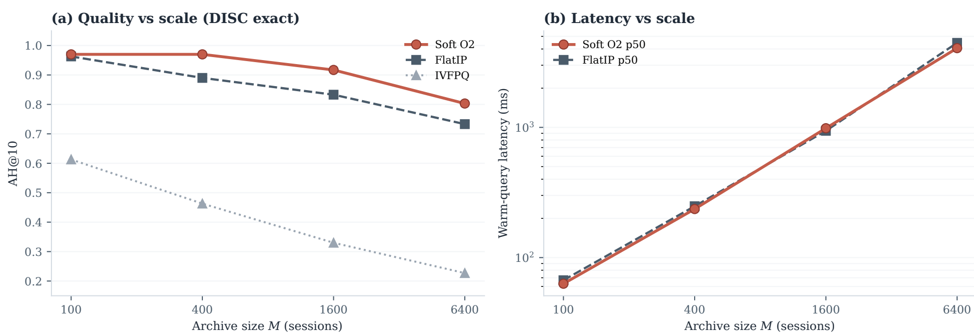

Figure 5: Scale curve on DISC-Law exact. Soft O2 keeps an AH edge over FlatIP as M grows; warm-query latency scales with archive size; IVFPQ quality collapses at moderate M.

incomplete retrievals into complete ones without shrinking session-miss rates (Table 4). O1 restores score fidelity; Soft O2 supplies the main gain through soft sibling inheritance; O3 makes large shared stores tractable. The gap-ratio analysis (Section 3.4) motivates keeping β close to but below unity; the AH sweep peaks in $\{0.9, 0.95, 0.98\}$ rather than at hard hydration (β=1). On CAIL, Soft O2 raises Answer Hit by large margins (U1 +0.218, Uk +0.428, U-last +0.517) with collapsing shuffled-*sid* controls, and keeps higher EC than hard expansion despite occasional AH ties. After Holm adjustment on the Soft O2 vs FlatIP family, the non-exact CAIL and LegalEp gains remain significant, while LegalEp exact replay is mixed (Table A.2). LegalEp paraphrase and advice-recall, rather than exact replay, support the main transfer result. Graded nDCG agrees with the Soft O2>FlatIP ordering (Table 6). Under Mix, gated Hybrid recovers additional cross-session gold on LegalEp-Lawyer with a directional lift on LegalMem-MT (Table 8); Soft O2 remains preferable when gold already shares *sid* (Table A.1). Turn-level BM25, RRF, and CE place Soft O2 among standard IR alternatives Robertson and Zaragoza (2009); Cormack et al. (2009); Nogueira et al. (2019); joint-episode indexes are a coarser design choice outside this protocol.

6.2. RAG application perspective

Consultation-memory retrieval is judged downstream by whether an LLM can answer from recovered evidence rather than confabulate. We audit this end-to-end with a dual-protocol pipeline on Soft O2 retrieval only: generator `qwen3:14b` and independent judge `qwen3:32b` are distinct models. Per corpus we pool N=270 judged instances as P1 n=120 plus P2 n=150 (folder `legal_qa_n270`). The independent judge assigns a continuous evidence-answerability score in $[0, 1]$ from the retrieved context and the generated reply; Evidence-Answerability Rate (EAR) is the fraction of instances with judge score ≥ 0.7. Table 9 reports pooled EAR and the mean judge score on a stratified subsample (n=90/corpus; 30 each of exact, paraphrase, and follow-up), labelled "Stratified judge" in the table. In the complete-episode stratum—both question and answer turns in top-k—judged answerability is higher in this single-system audit (pooled EAR 0.867 on DISC and 0.859 on Lawyer; Wilson 95% CI $[0.83, 0.90]$ / $[0.82, 0.89]$). Follow-up queries remain the hardest stratum (independent-judge EAR 0.517 / 0.480), consistent with episodic incompleteness persisting even when surface-form overlap is low.

A blind two-annotator legal audit (n=60/corpus) separates retrieval failures (gold episode absent from top-k) from generation failures given retrieved evidence. Annotators used a binary answerability label on each instance; agreement is Cohen's κ computed on those paired labels (0.71 on DISC, 0.68 on Lawyer). Retrieval-failure share is 0.283 / 0.317. Wrong-session contamination rates stay low (0.083 / 0.100), suggesting that complete episode retrieval under Soft O2 is associated with lower rates of a common RAG failure mode in which the model answers from a neighbouring client's matter. Hallucinated-legal-basis rates (0.117 / 0.133) remain non-zero, indicating that grounding in retrieved episodes does not guarantee statutory correctness—a scope boundary for deployment. AH and EAR measure evidence availability and judged answerability respectively; statutory correctness, citation validity, and professional liability lie outside this evaluation.

Table 9: End-to-end QA audit (N=270 per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`). Pooled EAR: fraction with judge score ≥ 0.7. Stratified judge: mean judge score on n=90/corpus (exact/paraphrase/follow-up). Human κ: Cohen's κ on binary answerability labels from two blind annotators (n=60/corpus).

Corpus	**Pooled EAR**	**Stratified judge**	**Human κ**
DISC-Law	**0.867**	0.757	0.71
Lawyer-LLaMA	**0.859**	0.729	0.68

6.3. Limitations

LegalEp episodes are consultation pairs; multi-turn authenticity remains reserved for CAIL.

Dependency on session segmentation and cluster quality.. Soft O2 relies on correct session identifiers (*sid*). In uncurated client logs, boundary noise—mid-session topic shifts, multi-intent interruptions, or automated segmentation drift—can corrupt sibling sets. If unrelated turns share a *sid*, attenuated inheritance may propagate false context; if a true advice turn is split into another session, Soft O2 cannot bind it. Soft O2-C likewise depends on BIRCH cluster purity. Under the Mix protocol, Sa/Sb pairs receive a shared cluster id so binding can be tested when thematic co-membership is present (Section 4.3); in fully unsupervised consolidation, noisier boundaries shrink the margin over dense retrieval. Deployments on raw consultation streams therefore need reliable upstream session segmentation, and Soft O2-C should be treated as gated rather than always-on.

Joint-episode indexing.. Concatenating full question–answer pairs into episode documents prevents incompleteness by changing the evidence unit from turns to merged documents (Section 2). That design is outside the Soft O2 turn-binding protocol; a matched comparison under online updates, latency, and fine-grained advice isolation is left to future work.

Embedding model and other scope limits.. All main grids use `nomic-embed-text-v2-moe`. A Chinese-specialized encoder such as `bge-large-zh-v1.5` could alter absolute dense ranks and the measured ρ distribution, and might shrink or widen Soft O2's relative gain wherever the bi-encoder already places answer turns inside top-k. We did not re-run the full Soft O2 grids under alternative encoders; the binding operator itself is encoder-agnostic once fused scores are available. The ρ statistics use an exact-channel query proxy and the strongest measured distractor; paraphrase and advice-recall gap laws may shift the feasible band slightly, though the AH sweep keeps a stable high-β region on those channels. Main $M{\approx}3000$ tables use a fixed seed; five-seed variance is reported only for the smaller $M{=}400$ store. LLM paraphrases need human audit. Statutory correctness, citation validity, and professional liability remain unevaluated. Because many channels are tested, Holm-adjusted p-values accompany the Soft O2 vs FlatIP family (Table A.2); Hybrid Mix gains outside LegalEp-Lawyer are treated as directional.

Latency and deployment scope.. Scale curves cover $M \in \{100, 400, 1600, 6400\}$ under a fixed hardware stack (Table A.3). At $M{=}6400$, warm-query p50 latency exceeds 4 s (4061 ms for Soft O2; 4458 ms for FlatIP)—unacceptable for millisecond interactive search. BIMS-LEGAL should therefore be positioned as **high-precision offline or quasi-real-time consultation memory recovery**—e.g., pre-meeting case review, audit replay, or batch RAG over a firm's advice archive—not as a sub-millisecond search engine. Future work will combine approximate nearest-neighbour indexes (e.g., HNSW) with Soft O2 and Soft O2-C binding layered on top of faithful candidate retrieval, preserving episode completion while reducing query latency. Behavior at roughly 10^5 sessions is not measured, and we do not claim asymptotic scalability from O3 wall-clock alone.

7. Conclusion

BIMS-LEGAL recovers prior advice from Chinese legal consultation archives under same-domain interference with a dual store: an episodic turn index and a BIRCH semantic store. Soft O2 is the main algorithmic contribution: attenuated session binding completes episodes without hard score copy. Soft O2-C and gated Hybrid extend the same idea to thematic clusters when Mix reconstructions place some gold off-session. Attenuation β is chosen by validation sweep under a three-candidate score-competition account; O1 FlatIP and O3 bulk-load are the implementation operators that make the shared-store ablations feasible. On CAIL multi-turn and LegalEp paraphrase /advice-recall channels, Soft O2 improves Answer Hit over FlatIP, keeps higher episode completeness than hard expansion on CAIL, and raises graded nDCG while cutting incompleteness; shuffled-*sid* controls collapse the gains. Under Mix, gated Hybrid recovers additional cross-session gold on LegalEp-Lawyer while Soft O2 remains preferable on intact same-session gold. An end-to-end Soft O2 QA audit associates complete episode retrieval with judged answerability and low wrong-session contamination, without claiming paired generation superiority over FlatIP. Turn-level BM25, RRF, and CE place Soft O2 among standard IR alternatives; joint-episode indexing remains a coarser design outside this protocol.

Declarations

Funding

Funding information omitted for anonymous review.

Conflict of interest

The authors declare no competing interests.

Ethics approval

This study uses publicly released research corpora and excludes confidential client data. The system is a research prototype for consultation-memory retrieval.

Data availability

Code and processed evaluation artifacts will be released upon publication. Upstream corpora remain under their original licences: CAIL2024 consultation tracks; Hugging Face `ShengbinYue/DISC-Law-SFT` and `Skepsun/lawyer_llama_data`. Processed LegalEp /LegalMem artifacts are derived research releases.

Code availability

Implementation artifacts and per-table JSON files will be released with the camera-ready version.

Author contributions

Author contributions omitted for anonymous review.

References

Ashley, K.D., 2017. Artificial Intelligence and Legal Analytics: New Tools for Law Practice in the Digital Age. Cambridge University Press.

Askari, A., Aliannejadi, M., Abolghasemi, A., Kanoulas, E., Verberne, S., 2024. Answer retrieval in legal community question answering, in: Advances in Information Retrieval (ECIR), Springer. pp. 477–485.

Atkinson, R.C., Shiffrin, R.M., 1971. The control of short-term memory. Scientific American 225, 82–91.

Callan, J.P., Lu, Z., Xu, S., 1994. Passage-level evidence in document retrieval, in: Proceedings of the 17th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 302–310.

Chen, J., Xiao, S., Zhang, P., Luo, K., Lian, D., Liu, Z., 2024. M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation, in: Findings of the association for computational linguistics: ACL 2024, pp. 2318–2335.

Cormack, G.V., Clarke, C.L.A., Buettcher, S., 2009. Reciprocal rank fusion outperforms condorcet and individual rank learning methods, in: Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 758–759.

Dai, Z., Xiong, C., Liu, Z., Callan, J., 2019. Context-aware sentence/passage encoding for answering factoid questions, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, pp. 241–251.

Gutiérrez, B.J., Shu, Y., Gu, Y., Yasunaga, M., Su, Y., 2024. Hipporag: Neurobiologically inspired long-term memory for large language models. Advances in neural information processing systems 37, 59532–59569.

Huang, Q., Tao, M., Zhang, C., An, Z., Jiang, C., Chen, Z., Wu, Z., Feng, Y., 2023. Lawyer llama technical report. `arXiv:2305.15062`.

Johnson, J., Douze, M., Jégou, H., 2019. Billion-scale similarity search with GPUs. IEEE Transactions on Big Data 7, 535–547.

Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., Yih, W.t., 2020. Dense passage retrieval for open-domain question answering, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 6769–6781.

Kumaran, D., Hassabis, D., McClelland, J.L., 2016. What learning systems do intelligent agents need? complementary learning systems theory updated. Trends in cognitive sciences 20, 512–534.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.t., Rocktäschel, T., et al., 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. Advances in neural information processing systems 33, 9459–9474.

Liu, B., Wu, Y., Zhang, F., Liu, Y., Wang, Z., Li, C., Zhang, M., Ma, S., 2022. Query generation and buffer mechanism: Towards a better conversational agent for legal case retrieval. Information Processing & Management 59, 103051.

Louis, A., van Dijck, G., Spanakis, G., 2024. Interpretable long-form legal question answering with retrieval-augmented large language models, in: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 22266–22275.

Maharana, A., Lee, D.H., Tulyakov, S., Bansal, M., Barbieri, F., Fang, Y., 2024. Evaluating very long-term conversational memory of llm agents, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 13851–13870.

Martinez-Gil, J., 2023. A survey on legal question–answering systems. Computer Science Review 48, 100552.

McClelland, J.L., McNaughton, B.L., O'Reilly, R.C., 1995. Why there are complementary learning systems in the hippocampus and neocortex: insights from the successes and failures of connectionist models of learning and memory. Psychological review 102, 419.

Mo, F., Gao, Y., Wu, Z., Liu, X., Chen, P., Li, Z., Wang, Z., Li, X., Jiang, M., Nie, J.Y., 2026. Leveraging historical information to boost retrieval-augmented generation in conversations. Information Processing & Management 63, 104449.

Modarressi, A., Imani, A., Fayyaz, M., Schütze, H., 2023. Ret-llm: Towards a general read-write memory for large language models. arXiv preprint arXiv:2305.14322 .

Nogueira, R., Yang, W., Cho, K., Lin, J., 2019. Multi-stage document ranking with BERT. `arXiv:1910.14497`.

Nussbaum, Z., et al., 2025. Nomic embed multimodal: Interleaved text, image, and datasets for general-purpose vision-language embeddings. Model card: `nomic-embed-text-v2-moe` (Nomic AI). Embedding model used for turn encoding in this work.

Packer, C., Wooders, S., Lin, K., Fang, V., Patil, S.G., Stoica, I., Gonzalez, J.E., 2023. Memgpt: Towards llms as operating systems. arXiv preprint arXiv:2310.08560 .

Rasmussen, P., Paliychuk, P., Beauvais, T., Ryan, J., Chalef, D., 2025. Zep: A temporal knowledge graph architecture for agent memory. arXiv preprint arXiv:2501.13956 .

Robertson, S., Zaragoza, H., 2009. The probabilistic relevance framework: BM25 and beyond. Foundations and Trends in Information Retrieval 3, 333–389.

Tan, H., Zhang, Z., Ma, C., Chen, X., Dai, Q., Dong, Z., 2025. Membench: Towards more comprehensive evaluation on the memory of llm-based agents, in: Findings of the Association for Computational Linguistics: ACL 2025, pp. 19336–19352.

Tulving, E., et al., 1972. Episodic and semantic memory. Organization of memory 1, 1.

Wang, Y., Gao, Y., Chen, X., Jiang, H., Li, S., Yang, J., Yin, Q., Li, Z., Li, X., Yin, B., et al., 2024. Memoryllm: Towards self-updatable large language models. arXiv preprint arXiv:2402.04624 .

Wu, D., Wang, H., Yu, W., Zhang, Y., Chang, K.W., Yu, D., 2024. Longmemeval: Benchmarking chat assistants on long-term interactive memory. arXiv preprint arXiv:2410.10813 .

Wu, Y., Liang, S., Zhang, C., Wang, Y., Zhang, Y., Guo, H., Tang, R., Liu, Y., 2025. From human memory to ai memory: A survey on memory mechanisms in the era of llms. `arXiv:2504.15965`.

Yue, S., Chen, W., Wang, S., Li, B., Shen, C., Liu, S., Zhou, Y., Xiao, Y., Yun, S., Huang, X., Wei, Z., 2023. Disc-lawllm: Fine-tuning large language models for intelligent legal services. `arXiv:2309.11325`.

Yue, S., Liu, S., Zhou, Y., Shen, C., Wang, S., Xiao, Y., Li, B., Yun, S., Shen, X., Chen, W., et al., 2024. Lawllm: Intelligent legal system with legal reasoning and verifiable retrieval, in: International Conference on Database Systems for Advanced Applications, Springer. pp. 304–321.

Zaheer, M., Kottur, S., Ravanbakhsh, S., Poczos, B., Salakhutdinov, R., Smola, A.J., 2017. Deep sets, in: Advances in Neural Information Processing Systems.

Zhang, T., Ramakrishnan, R., Livny, M., 1996. BIRCH: an efficient data clustering method for very large databases, in: Proceedings of the 1996 ACM SIGMOD International Conference on Management of Data, pp. 103–114.

Zhang, Y., Zhai, S., Meng, Y., Bi, S., Chen, Y., Qi, G., 2024. Event is more valuable than you think: Improving the similar legal case retrieval via event knowledge. Information Processing & Management 61, 103729.

Zhong, W., Guo, L., Gao, Q., Ye, H., Wang, Y., 2024. Memorybank: Enhancing large language models with long-term memory, in: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 19724–19731.

Appendix A. Numeric grids and controls

Same-session Soft O2-C contrasts, significance tests, scale curves, and full Soft O2 numeric grids appear below.

Table A.1: Same-session Soft O2 vs ungated Soft O2-C (AH@10). Soft O2 is preferable when gold already shares *sid* with the query; ungated Soft O2-C can surface thematic confusers. Boldface: better AH within each row. Mix Soft O2-C results appear in Table 8.

Corpus / channel	**Soft O2**	**Soft O2-C**
LegalEp-DISC / exact	**0.836**	0.744
LegalEp-DISC / u-para	**0.807**	0.730
LegalEp-DISC / advice	**0.548**	0.490
LegalEp-Lawyer / exact	**0.818**	0.710
LegalEp-Lawyer / u-para	**0.817**	0.733
LegalEp-Lawyer / advice	**0.546**	0.448
LegalMem-MT / U1	**0.903**	0.712
LegalMem-MT / Uk	**0.793**	0.385

Table A.2: McNemar mid-p on paired Answer Hit (Soft O2 vs FlatIP / hard hydration / FlatIP+CE). Holm-adjusted p is reported for the nine primary Soft O2 vs FlatIP contrasts.

Corpus / channel	**Contrast**	**ΔAH**	**mid-p**	**Holm p**
CAIL / Uk	O2 vs FlatIP	+0.428	8.99e-48	8.09e-47
CAIL / Uk	O2 vs Hard hydr.	-0.022	0.243	—
CAIL / U1	O2 vs FlatIP	+0.218	5.19e-18	4.67e-17
CAIL / U1	O2 vs Hard hydr.	-0.013	0.435	—
CAIL / U-last	O2 vs FlatIP	+0.517	5.51e-65	4.96e-64
CAIL / U-last	O2 vs Hard hydr.	-0.028	0.0982	—
CAIL / Uk	O2 vs FlatIP+CE	+0.323	2.50e-26	—
CAIL / U1	O2 vs FlatIP+CE	+0.100	3.51e-05	—
CAIL / U-last	O2 vs FlatIP+CE	+0.420	3.92e-41	—
LegalEp-DISC / exact	O2 vs FlatIP	-0.002	0.934	0.934
LegalEp-DISC / exact	O2 vs Hard hydr.	+0.024	0.302	—
LegalEp-Lawyer / exact	O2 vs FlatIP	+0.060	0.0239	0.191
LegalEp-Lawyer / exact	O2 vs Hard hydr.	+0.140	3.50e-10	—
LegalEp-DISC / u-para	O2 vs FlatIP	+0.101	7.95e-05	6.36e-04
LegalEp-DISC / u-para	O2 vs Hard hydr.	+0.056	0.035	—
LegalEp-Lawyer / u-para	O2 vs FlatIP	+0.191	2.10e-15	1.68e-14
LegalEp-Lawyer / u-para	O2 vs Hard hydr.	+0.163	7.57e-09	—
LegalEp-DISC / advice	O2 vs FlatIP	+0.086	1.25e-04	8.75e-04
LegalEp-DISC / advice	O2 vs Hard hydr.	+0.026	0.215	—
LegalEp-Lawyer / advice	O2 vs FlatIP	+0.140	1.18e-10	9.44e-10
LegalEp-Lawyer / advice	O2 vs Hard hydr.	+0.078	5.72e-04	—

Table A.3: Scale curve on DISC-Law (mean over 3 repeats). Latency is warm-query p50/p95. Boldface: best AH@10 within each M block; lowest p50/p95 within each M block.

M	**Mode**	**AH@10**	**Build (s)**	**p50 (ms)**	**p95 (ms)**
100	FlatIP	0.963	3.0	67	80
100	IVFPQ	0.613	**25.2**	**40**	**54**
100	Soft O2	**0.970**	3.2	63	69
400	FlatIP	0.890	26.5	248	305
400	IVFPQ	0.463	89.5	**171**	**204**
400	Soft O2	**0.970**	30.9	236	313
1600	FlatIP	0.833	359.8	**941**	1278
1600	IVFPQ	0.330	**251.4**	669	**751**
1600	Soft O2	**0.917**	393.7	985	1172
6400	FlatIP	0.733	**4722.8**	4458	4969
6400	IVFPQ	0.227	1072.1	**2182**	**3021**
6400	Soft O2	**0.803**	5245.7	4061	4503

Table A.4: Soft O2 AH@10 versus β on the primary shared-corpus evaluation. Peak performance lies in $\{0.9, 0.95, 0.98\}$; aggressive attenuation (β=0.5) weakens binding.

Corpus	0.5	0.7	0.9	0.95	0.98	1.0
CAIL / Uk	0.323	0.537	0.697	0.700	0.698	0.662
LegalEp-Lawyer / exact	0.608	0.654	0.754	0.754	0.752	0.752
LegalEp-Lawyer / u-para	0.631	0.637	0.717	0.745	0.759	0.749
LegalEp-DISC / u-para	0.551	0.551	0.618	0.648	0.648	0.642

Table A.5: CAIL2024 shared-corpus results ($M \approx 3000$). Boldface marks the best AH@10 and best EC@10 within each channel (ties bolded jointly). **: Soft O2 vs FlatIP, McNemar mid-p<0.01. nDCG@10 uses gold-session IDCG.

Channel	System	AH@10	EC@10	nDCG@10	95% CI (AH)
U1	FlatIP	0.570	0.413	0.554	[0.530,0.610]
	Soft O2	0.788	**0.664**	0.776	[0.757,0.820]
	Hard hydr.	**0.802**	0.578	0.864	[0.768,0.835]
	Shuffled O2	0.452	0.363	0.411	[0.410,0.492]
Uk-followup	FlatIP	0.270	0.347	0.445	[0.237,0.307]
	Soft O2	0.698	**0.654**	0.793	[0.660,0.735]
	Hard hydr.	**0.720**	0.585	0.847	[0.685,0.753]
	Shuffled O2	0.230	0.320	0.369	[0.198,0.262]
U-last	FlatIP	0.245	0.335	0.420	[0.212,0.280]
	Soft O2	0.762	**0.659**	0.791	[0.725,0.797]
	Hard hydr.	**0.790**	0.586	0.854	[0.757,0.822]
	Shuffled O2	0.200	0.315	0.348	[0.172,0.232]

Soft O2 leads EC; hard hydration can lead AH.

Table A.6: LegalEp-DISC (M=3000, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * p<0.05, ** p<0.01 (Soft O2 vs FlatIP, McNemar mid-p). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCG.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	**0.640**	0.741	0.730
	Soft O2	0.638	**0.780**	0.813
	Hard hydr.	0.614	0.598	0.886
	Shuffled O2	0.448	0.645	0.666
advice-recall	FlatIP	0.342	0.438	0.466
	Soft O2	**0.428**	**0.503**	0.524
	Hard hydr.	0.402	0.405	0.572
	Shuffled O2	0.310	0.421	0.436
u-para	FlatIP	0.547	0.681	0.681
	Soft O2	**0.648**	**0.735**	0.759
	Hard hydr.	0.592	0.571	0.827
	Shuffled O2	0.469	0.610	0.627

Table A.7: LegalEp-Lawyer (M=3000, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. $^{*}p$<0.05, $^{**}p$<0.01 (Soft O2 vs FlatIP). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCG.

Channel	**System**	**AH@10**	**EC@10**	**nDCG@10**
exact	FlatIP	0.692	0.806	0.753
	Soft O2	**0.752**	**0.848**	0.869
	Hard hydr.	0.612	0.615	0.897
	Shuffled O2	0.570	0.747	0.705
advice-recall	FlatIP	0.332	0.464	0.483
	Soft O2	**0.472**	**0.540**	0.582
	Hard hydr.	0.394	0.403	0.601
	Shuffled O2	0.348	0.467	0.454
u-para	FlatIP	0.568	0.718	0.744
	Soft O2	**0.759**	**0.819**	0.844
	Hard hydr.	0.596	0.594	0.864
	Shuffled O2	0.592	0.712	0.705

Table A.8: Turn-level BM25 and dense+BM25 RRF (AH@10). Boldface: higher AH within each row among reported turn-level cells. BM25-joint / joint-episode document indexes are omitted: they change evidence granularity and were not run as Soft O2 protocol competitors.

Corpus	**Channel**	**AH@10**
LegalEp-DISC	exact (BM25-turn)	0.802
LegalEp-DISC	advice-recall (BM25-turn)	0.382
LegalEp-DISC	exact (dense+BM25 RRF)	0.780
LegalEp-DISC	advice (dense+BM25 RRF)	0.394
CAIL	U1 / Uk / U-last (BM25-turn)	0.748 / 0.482 / 0.413
LegalEp-Lawyer	exact / advice (BM25-turn)	0.784 / 0.460